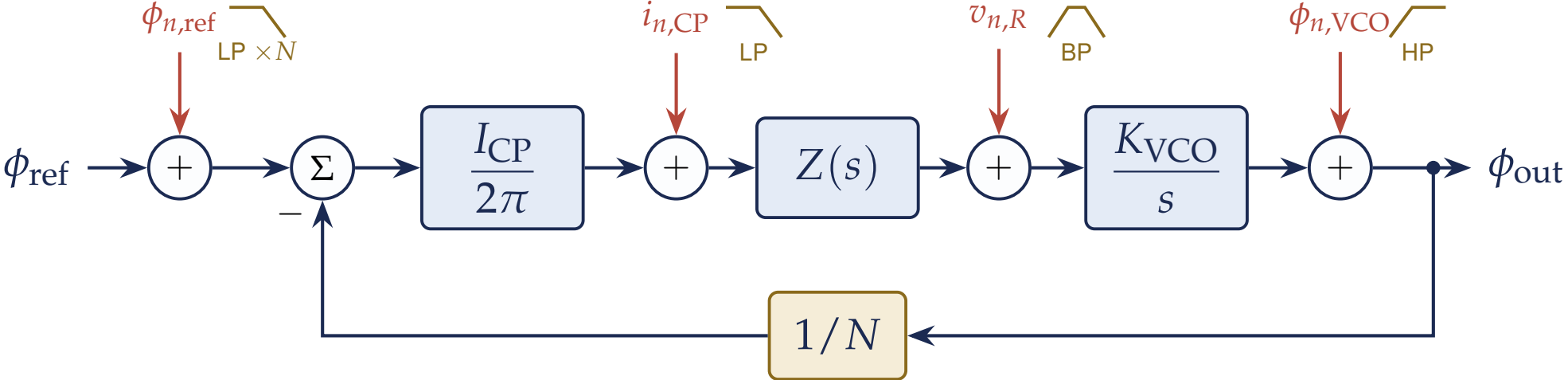

$\phi_{n,\mathrm{ref}}$
LP $\times N$
$i_{n,\mathrm{CP}}$
LP
$v_{n,R}$
BP
$\phi_{n,\mathrm{VCO}}$
HP
ϕ_{ref}
+
Σ
−
$\frac{I_{\mathrm{CP}}}{2\pi}$
+
$Z(s)$
+
$\frac{K_{\mathrm{VCO}}}{s}$
+
ϕ_{out}
$1/N$